Brent Vine

Gk. *σφήν*, Eng. *spoon*: A Note on »Eichner's Law«

1. Greek *σφήν*, gen. *σφηνός* m. 'wedge' is first attested in Aeschylus and Aristophanes, in both cases outside lyric passages, and is otherwise restricted to Attic and Ionic (see the detailed survey of attestations in Solmsen 1909: 129). Without further information, then, we cannot tell whether the form derives from a Pr.-Gk. **sp^{h}ḗn* or **sp^{h}ā́n*.

By a stroke of good fortune, further information is available, in the form of an early Central Ionic inscription, from a time (5th c.) and place (Ceos) in which *ē* (spelled <E>) and *ā* (spelled <H>) are still regularly kept apart: as seen by Schulze (1913: 190f. = 1933: 379), the spelling ΣΦΗΝΟΠΟ[Δ]Ι (in the phrase ΕΓ ΚΛΙΝΗΙ ΣΦΗΝΟΠΟ[Δ]Ι 'on a bier with wedge-shaped legs'; IG XII.5.593 A.6 = Schwyzer *Del.* 766, Buck 8) guarantees a source **sp^{h}ā́n*, as is now generally assumed, no doubt correctly (thus e.g. Schwyzer 1939: 487n7, Frisk GEW s.v. *σφήν*). As Schulze pointed out, moreover, *a*-vocalism of some sort is also favored by the gloss *ἐν σφανίῳ· ἐν κλιναρίῳ* Hsch., for which Schulze reasonably supposed that *σφανίον* (or *σφάνιον*) is »eine Art substantivierter Kurzform für *σφηνόπους*«.[1]

What, then, is the correct historical interpretation of this **sp^{h}ā́n*? There appear to be two main possibilities, as pointed out by Frisk *loc. cit.*, namely (a) interpretation as a root noun, thus **sp^{h}áns(–)* (cf. *χήν* 'goose' < **ĝ^{h}ans-*), or (b) as an *n*-stem, roughly **sp^{h}a-ḗn* (cf. Chantraine 1933: 166f.). The second of these is distinctly favored by a series of probably or possibly related words for 'blade, spear, spade' and the like, in Greek and elsewhere, including the following:

- *σπάθη* 'blade' (wood, metal, plants);
- Gmc. **spadan-* (NHG *Spaten*) and **spadō* (OE *spadu* etc.) 'spade';
- Ved. *sphyá-* [AV, TS+], a »sword-like« sacrificial implement (esp. for stirring liquids, or for delimiting sacred space), later (Br.+) 'boom, spar'.

[1] Similarly LSJ (s.v. *σφάνιον*); likewise Wahrmann (1914: 163), apparently writing without knowledge of Schulze's analysis.

Münchener Studien zur Sprachwissenschaft, 62, 2002, S. 289–299
[2006]

In general, the most economical assumption for all of these would involve a root-shape **speh₂-*: thus **sph₂-dʰ-* in σπάθη, Gmc. **spad-*,[2] and **sph₂-iio-* in Ved. *sphyá-*.[3] In this case, σφήν would continue an *n*-stem **sp(e)h₂-én*, and the Hesychian σφανίον (which may reflect either σφᾶν- or σφᾰ́ν-) could continue either a Doric **spʰā́n-* or else (in the case of σφᾰ́ν-) a form based on the zero grade allomorph **sph₂-n-*.[4] Also generally placed here (e.g. IEW 989, MALLORY/ADAMS 1997: 431) is a series of Italic and Celtic forms with the shape /spond-/, such as Lat. *sponda* 'bedstead, bed board', OIr. *sonn* 'staff, support', *sonnach* 'fence'. While such forms do not directly confirm or disfavor the assumption of a root-shape **speh₂-*, they could in principle help support the assumption of an *n*-stem, via formulations like **sph₂-on+d(ʰ)-* (or conceivably **sp(e)h₂-ōn+d(ʰ)-*, with Osthoff shortening).

2. At least since a very early discussion by A. KUHN (1855: 15), Greek σφήν 'wedge' has been plausibly connected, at least semantically, with a series of Germanic words denoting sharp oblong objects, especially the widespread terms for 'wood chip, shaving, sliver' represented by OE *spōn* (Mod. Eng. *spoon*), OHG *spān* (NHG *Span*), ON *spānn*. There is, however, a serious phonological difficulty with this comparison, namely the fact that these forms reflect Gmc. **spēnu-*, with *ē*-grade, which on its face is incompatible with the Pr.-Gk. **spʰā́n-* arrived at by SCHULZE (and now

[2] A **spn̥dʰ-* (cf. the possibility of Pr.-Gk. **spʰánts(–)* noted above) is technically possible for σπάθη, but will not work for the Gmc. material.

[3] Explicitly in terms of **sp(e)h₂-(dʰ-)*: NORMIER 1977: 173n6, 1981: 24n21.

[4] Despite the tendency of Greek to generalize unaspirated stops in forms deriving from a sequence **Th₂V* and despite the strictures of P. ELBOURNE (in a series of articles, most recently 2001) against the aspiration of voiceless stops by laryngeal, there is no real difficulty in assuming that the σφη- forms have generalized the aspirated stops resulting from original **sph₂V-* sequences (cf. also Ved. *sphyá-* above, not to mention οἶσθα), although the point cannot be pursued here. Another traditional comparandum within this set (see IEW 980) is Hitt. *išpā́tar* n., a term (especially common in ritual texts) for an implement used in eating and in removing things from a hearth; for the comparison of Hitt. *išpā́tar* with σπάθη, Ved. *sphyá-* and the Gmc. material, see more recently NORMIER, *locc. cit.* It remains unclear, however, whether Hitt. *išpā́tar* (the scriptio plena is assured) can derive from an original **spéh₂-tr̥*, which might instead have been expected to yield ***išpattar* or ***išpaḫtar*. See, however, HAJNAL 2001: 215n55 on secondary formations of the type OHitt. *antuḫšā́tar* 'Bevölkerung' (based on abstracts in **-ā* < **-ah₂* plus secondary **-tr̥/-tn-*).

generally assumed) for Gk. σφήν. Not surprisingly, this conundrum has elicited the following three responses:

· one can reject the connection altogether (this was the conclusion of SCHULZE's analysis, cf. FRISK *loc. cit.*: »Dadurch [*scil.* given the »Grundform **σφᾰ́ν*«] wird die herkömmliche ... Gleichsetzung mit einem germ. Wort für 'Span' ... hinfällig«);

· one can set up a root **speh₁-*, which will directly account for the Germanic forms (cf. IEW 980 *sp(h)ē-*, similarly WATKINS 2000: 82[5]) – but in that case, if one admits the connection of σφήν with the Germanic forms (as in fact is made explicit by both POKORNY and WATKINS), σφήν itself remains unexplained;

· one can set up **speh₂-*, as is done, for example, by A. DELLA VOLPE (*apud* MALLORY/ADAMS *loc. cit.*), on the strength of σφήν and the other forms with *a*-vocalism listed in §1. above (σπάθη etc.) – but in that case, the Germanic forms remain unaccounted for, as DELLA VOLPE herself pointedly notes: »the Gmc. forms as if < **spēn-* whose vowel is not well-explained«.

A more satisfying alternative suggests itself, namely that the original root shape is indeed **speh₂-* (as suggested by forms like σφήν, σπάθη, etc.), but that Gmc. »**spēnu-*« reflects a lengthened-grade **spēh₂nu-*, via the laryngeal treatment or behavior now widely known as »Eichner's Law«,[6] according to which PIE **ē* adjacent to **h₂* (and perhaps also **h₃*) does not undergo coloring. (See further §3. below for more general discussion of the process.)

The attractiveness of this analysis is to some extent independent of whatever morphological interpretation is given to the hypothesized **spēh₂nu-*, which for various reasons must remain somewhat unclear – not least of all because we do not know the precise meaning of the presumed underlying root **speh₂-*. One could speculate, for example, that **spēh₂nu-* is simply a primary »*nu*-stem« (thus **spéh₂-nu-*), of the type Ved. *bhā́nu-*, Av. *bānu-* 'light', etc. (WD II.2.741ff., BRUGMANN 1906:

[5] In the case of WATKINS 2000, this choice is eminently reasonable, given the emphasis of that work on English etymology.

[6] EICHNER 1973, cf. MAYRHOFER 1986: 132ff., 141f., MEIER-BRÜGGER 2000: 110.

290ff.); in that case, the lengthened grade of **spéh$_2$-nu-* could be accounted for by claiming that the formation was associated with an acrostatic present **spéh$_2$-ti/*sph$_2$-n̥ti* (which, of course, is purely hypothetical, such a form being unattested). Somewhat less arbitrary, especially given evidence for *n*-stem inflection elsewhere (cf. above on *σφήν* and on the Italic and Celtic /spond-/ forms), would be an interpretation **spéh$_2$-n-(u-)*. Thus, one could begin with a form **speh$_2$-(e)n-* meaning 'wedge, sharp point' or the like (cf. *σφήν*), which then serves as the basis for a thematic vr̥ddhi-derivative **spēh$_2$-n-ó-* 'having a sharp point' (cf. the Gmc. words for 'chip, sliver, etc.', which are readily interpretable in this way). In this event, one would only need to account for the development of *u*-stem inflection, and here it is suggestive that **-no- ~ *-nu-* alternations are independently documented (Att. *θρᾶνος* m. 'bench' : Ion. *θρῆνυς*, Myc. *ta-ra-nu/ta-ra-nu-we* 'footstool'; RV *vadhasná-* 'projectile' : *vadhasnu-* '[one who is] ready to strike; marksman'), including Germanic material (Ved. *tr̥ṇa-* n. 'grass, straw' : Go. *paurnus*, OE *porn*, etc., 'thornplant, thorn', cf. within Germanic Go. *-qairnus* 'mill' beside OSax. *quern*, OHG *chwirna* etc. 'millstone, *quern*', with **-nā-*).

However this may be – that is, even if one cannot insist on this or any other particular morphological interpretation for **spēh$_2$nu-* – the essential point remains that the semantically attractive and widely accepted (though phonologically hitherto obscure) etymological connection between Gk. *σφήν* and OE *spōn* etc. can be neatly rescued by an appeal to Eichner's Law, thereby contributing another item to the dossier of forms that may display this phenomenon; cf. the call issued by J. RASMUSSEN (1990/91: 400): »I strongly appeal to fellow researchers to look for additional examples [*scil.* of Eichner's Law]«.

3. In the paper just cited, RASMUSSEN attempted to address a series of criticisms leveled against Eichner's Law by F. O. LINDEMAN (1987: 56ff.). As RASMUSSEN noted, somewhat colorfully (1990/91: 396), »Lindeman has really taken the bull by the horns ...«; LINDEMAN's criticisms deserved to be taken seriously, and RASMUSSEN did so. In the end, he found none of the criticisms theoretically valid, and arrived at no less than twenty-five »highly probable PIE examples of uncoloured /ē/ in the vicinity of /h$_2$/«. LINDEMAN, however, remained unconvinced: he subsequently devoted an article to the problem (1994), and has trenchantly repeated his objections in his most recent survey of laryngeal theory (1997: 80ff.). It thus seems appropriate to append some remarks on Eichner's Law, with particular reference to the new typological argument presented in LINDEMAN's 1994 article and repeated in his later treatment.[7]

In his original presentation, EICHNER outlined the basic theoretical issue as follows:

> Die Annahme der Erhaltung von uridg. *ē* in dieser Position ist prinzipiell unbedenklich, da Langvokale erfahrungsgemäß durch benachbarte Konsonanten nicht in demselben Maß verändert werden wie die entsprechenden Kurzvokale. (1973: 72, cited also by LINDEMAN 1994: 110)

In response, LINDEMAN sees »no general phonologic justification for this claim«, noting that »various Semitic languages offer incontrovertible evidence for colouring of long vowels by adjacent laryngeal/pharyngeal consonants« (1994: 110–111). Since, he continues, at least some of the consonants denoted by PIE **h$_2$* and **h$_3$* may well have been of the laryngeal/pharyngeal type, »it follows that there is no phonologic reason *a priori* why these consonants should not have coloured an immediately preceding or following [ē] to [ā] and to [ō], respectively, in the parent Indo-European language« (1994: 111, cf. 1997: 80–81).

But as a typological argument, these observations do nothing to undermine Eichner's Law. They merely show that the »non-coloring« behavior proposed by EICHNER for PIE is not a valid phonological universal (a point about which EICHNER himself was silent), since long vowels adjacent to laryngeals and pharyngeals are in fact colored in Semitic. In fact, it is precisely to be expected that »coloring« effects of this sort will vary in this way from language to language, because of the phonetic nature of the process itself; in other words, there is good reason, *a priori*, to expect that such consonants might not have colored an adjacent long vowel in some languages, and might have done so in others. This is

[7] There is no question of attempting to address, in the compass of a short article, the various objections LINDEMAN raises to individual instances of Eichner's Law, many of which have in any event been discussed by RASMUSSEN (1990/91).

because the phonetic assimilations in question fundamentally involve, as EICHNER's formulation already implied, the timing of transitional gestures relative to the onset of a following segment; in the particular context at issue here, we are concerned with the transition of first and second formants at vowel offset, relative to the onset of a following post-velar obstruent. And in the case of a long vowel, it is perfectly natural that the transitional gestures might not spread over the entire duration of the vowel, in which case the original vowel quality remains unaffected for some portion of the duration of the vowel. Thus, from the point of view of typological argumentation, it is hardly sufficient to restrict the focus to Semitic, where, as it happens, the timings in question merely operate in one particular way; one must also show that these timings are also operative in all other known cases of systems with post-velar obstruents (and concomitant »coloring« effects) typologically similar to that reconstructed for Proto-Indo-European.

Among such systems, it is the Salishan languages of the Pacific Northwest that provide the most suggestive and potentially probative parallels to PIE.[8] In many of these languages, vowels in the vicinity of pharyngeals and uvulars are »retracted« or show »faucal harmony« (the standard terms in Salishan linguistics; see CZAYKOWSKA-HIGGINS/KINKADE 1998b: 12, BESSELL 1998, MITHUN 1999: 27f.); or, as the phenomenon is sometimes described, vocalic oppositions in these environments are neutralized in favor of vocalic variants characterized by lowering and/or backing of the tongue root.[9] Consider, for example, the neutralization of /e/ as [a] in Shuswap before uvular obstruents and before or after uvularized resonants (KUIPERS 1974: 22) – thus in /sχélwe/[10] 'husband', phonetically [sχælwɛ], the reduplicating possessed form places the base form's /e/ of the first syllable directly before the uvular fricative, yielding /γ-n-sχéχlwe/, phonetically [γɛ͡nsχáχælwɛ] (KUIPERS 1974: 27).[11] In PIE terminology, /e/ is »colored« to /a/ before the uvular fricative (cf. PIE $**eh_2 > *ah_2$). Interestingly, KUIPERS opts to write /e/ in such cases »for reasons of regularity of structure and simplicity of rules« (1974: 22), precisely corresponding to the widespread use, in IE studies, of idealized »structural« notations like $*eh_2$ (vs. $*ah_2$), $*h_3e$ (vs. $*h_3o$), and the like.

Salishan phonological inventories do not commonly include long vowels, but these are assured in some languages, as for example in Bella Coola,[12] which includes /ī/ (phonetically [e:]), /ū/, and /ā/ beside /i/, /u/, and /a/. And here we find such items as /cīχ/ 'dig', /ʔīχ^{W}/ 'burn', and /–χlq′īχ^{W}-/[13] 'turn around', in which the long front vowel (in Bella Coola generally realized, phonetically, as a mid vowel) is *not* described as retracted (or »colored«) to low/back position in the environment before a directly following uvular or labio-uvular fricative. (For the latter configuration, cf. PIE $*h_3$ and several possible Eichner's Law cases involving $*ēh_3$: Hitt. *ganēšš-* < $*ĝnéh_3$*-s-*, etc.; see RASMUSSEN 1990/91: 408, with further references).

There is much about Eichner's Law that continues to be debated, such as questions about the range and extent of the phenomenon, or about its chronology,[14] not to mention new proposals for additional instances, such as that involving Gk. *σφήν* and Gmc. **spēnu-* discussed in §§1. and 2. above.[15] There are, however, no compelling phonetic or typological arguments against Eichner's Law for PIE. Given the presumed phonetic nature of the process as one involving the relative timing of formant transitions, the data from Semitic adduced by LINDEMAN have no bearing on the question, while the Salishan data sketched above provide

[8] I am deeply grateful to Andrew GARRETT for bringing the Salishan data to my attention in this connection. For general orientation on Salishan phonological systems, see CZAYKOWSKA-HIGGINS/KINKADE 1998b: 7–22; specifically on post-velar phonology in Salishan: BESSELL 1992 (with extensive comparative discussion of Caucasian and Semitic).

[9] See especially Bessell 1998 for detailed spectrographic analysis, which reveals the F1 raising and F2 lowering typical of vocalic transitions in the environment before post-velars.

[10] /χ/ = voiceless uvular fricative.

[11] Among many other examples that could be adduced: similar behavior in Saanich (North Straits Salish), where /e/ is »often lax, lowered, and centralized in the environment of uvulars, labio-uvulars, and /ʔ/« (MONTLER 1986: 19).

[12] See fundamentally NEWMAN 1947: 131, and more recently NATER 1979: 179f., DAVIS/SAUNDERS 1997: 23n2, CZAYKOWSKA-HIGGINS/KINKADE 1998b: 10.

[13] /q′/ = uvular ejective stop.

[14] See recently e.g. NASSIVERA 2000: 58n6 (»Eichner's law did not operate in Baltic ..., Greek ..., Latin ... and probably anywhere outside Anatolian«), and for the question of chronology, MOTTAUSCH 2000: 85f. (Eichner's Law as one of several factors pointing to the relatively late development of lengthened-grade vocalism in PIE).

[15] For another recent proposal, see KIM 2000.

explicit typological support in favor of the behavior envisioned by EICHNER.[16]

Korrekturnachtrag

Ad §2.: Neri makes the attractive suggestion (per litt., July 2004) that Gmc. **spēnu-* could reflect an inner-Gmc. generalization of the apparent stem of acc. sg. **spēnu^n*, within the Gmc. paradigm of the inherited hysterokinetic **sph$_2$-én* (> Gmc. nom. **spēn*, gen. **spēniz*, acc. **spēnu^n* → »**spēnu-*«). (Note that in this event, an »Eichner's Law« context – though with the long vowel and the laryngeal in the reverse order from what was assumed in §2. – is still a necessary component of the derivational history.) Neri further points out that Central Ion. **sp^{h}ā́n* may itself have an analogical source, based on oblique **sp^{h}an-* (< acc. **sph$_2$-én-m̥*, gen. **sph$_2$-n-és*, etc.), vis-à-vis the presumed original nom. sg. **sp^{h}é́n*.

Ad §3.: For another example of »retraction« in Salish, note the lowering of /i/ to [ɛ] adjacent to post-velars and glottals in Sliammon (Mainland Comox); H. Watanabe, *A Morphological Description of Sliammon, Mainland Comox Salish, with a Sketch of Syntax* (Kyoto, 2003) 11. Similar behavior is also documented for other Pacific NW groups with comparable inventories, e.g. Coast Tsimshian, where morphophonemic {ə} is lowered to /a/ adjacent to a post-velar or glottal; F. Sasama, *A Descriptive Study of the Coast Tsimshian Morphology* (Ph.D. dissertation, Kyoto University, 2001) 34. Particularly interesting in connection with Eichner's Law are the Coast Tsimshian »interrupted vowels«, i.e. surface phonetic sequences [VʔV] as realizations of phonemic long vowels in the position before post-velars (Sasama 2001:15) – e.g., with long front vowel before glottalized uvular stop, /néːqʼ/ = [néʔɐχ] 'anal fin, dorsal fin' (the spirantization of the word-final post-velar is regular). Thus, as a striking parallel to Eichner's Law, the backing effect of the post-velar *does not extend throughout the entire duration of the preceding vowel*, with the result

[16] An earlier version of this paper was presented at the 20th East Coast Indo-European Conference (Cornell University, June 2001); I am indebted to the participants of the conference for their helpful comments and suggestions, in particular Andrew GARRETT, Jay JASANOFF, Ron KIM, Jared KLEIN, Craig MELCHERT, Birgit OLSEN, Jens RASMUSSEN, and Elisabeth RIEKEN.

that the first half of the long vowel remains mid front in terms of vowel quality. (I am grateful to K. Yoshida [Kyoto University] for bringing the work of Watanabe and Sasama to my attention.)

References

BESSELL, N. J. 1992. Towards a Phonetic and Phonological Typology of Post-Velar Articulation. Ph.D. dissertation, The University of British Columbia (Department of Linguistics).

–, 1998. Phonetic aspects of retraction in Interior Salish, in CZAYKOWSKA-HIGGINS/KINKADE 1998a: 125–152.

BRUGMANN, K. 1906. Grundriß der vergleichenden Grammatik der indogermanischen Sprachen^2, II–1. Straßburg, Trübner.

BUCK: BUCK, C. D. 1955. The Greek Dialects. University of Chicago Press.

CHANTRAINE, P. 1933. La formation des noms en grec ancien. Paris, Klincksieck.

CZAYKOWSKA-HIGGINS, E. and D. KINKADE (eds.). 1998a. Salish Languages and Linguistics: Theoretical and Descriptive Perspectives. Berlin/New York, De Gruyter (Trends in Linguistics, Studies and Monographs 107).

–, 1998b. Salish languages and linguistics, in CZAYKOWSKA-HIGGINS/KINKADE 1998a: 1–68.

DAVIS, P. W. and R. SAUNDERS. 1997. A Grammar of Bella Coola. Missoula, MT, University of Montana Occasional Papers in Linguistics No. 13.

EICHNER, H. 1973. Die Etymologie von heth. *meḫur*. MSS 31, 53–107.

ELBOURNE, P. 2001. Aspiration by /s/ and devoicing of mediae aspiratae. HS 114, 197–219.

FRISK GEW: FRISK, H. 1960–1972. Griechisches etymologisches Wörterbuch. Heidelberg, Winter.

HAJNAL, I. 2001. Silbenschrift und Silbenstruktur in den indogermanisch-anatolischen Sprachen Kleinasiens, in Lingua et Linguae: Festschrift für Clemens-Peter HERBERMANN zum 60. Geburtstag (ed. U. H. WASSNER), 189–224. Aachen, Shaker Verlag (BBS Neue Folge 6).

IEW: POKORNY, J. 1959. Indogermanisches etymologisches Wörterbuch. repr. Tübingen/Basel, Francke.

KIM, R. 2000. 'To drink' in Anatolian, Tocharian, and Proto-Indo-European. HS 113, 151–170.

KUHN, A. 1855. Über das alte S und einige damit verbundene Lautentwicklungen. Artikel VI, VII und Nachträge. KZ 4, 1–46.

KUIPERS, A. H. 1974. The Shuswap Language: Grammar, Texts, Dictionary. The Hague/Paris, Mouton.
LINDEMAN, F. O. 1987. Introduction to the 'Laryngeal Theory'. Oslo, Norwegian University Press.
–, 1994. 'Laryngeal' colouring and lengthened grade in Indo-European, in Indogermanica et Caucasica: Festschrift für Karl Horst SCHMIDT (ed. R. BIELMEIER and R. STEMPEL), 110–121. Berlin/New York, De Gruyter.
–, 1997. Introduction to the 'Laryngeal Theory'. Innsbruck, IBS 91.
LSJ: LIDDELL, H. G., R. SCOTT, H. S. JONES and R. MCKENZIE. 1940–1996. A Greek-English Lexicon. Oxford, Oxford University Press.
MALLORY, J. P. and D. Q. ADAMS. 1997. Encyclopedia of Indo-European Culture. London, Fitzroy and Dearborn.
MAYRHOFER, M. 1986. Indogermanische Grammatik, I/2 (Lautlehre). Heidelberg, Winter.
MEIER-BRÜGGER, M. 2000. Indogermanische Sprachwissenschaft. Berlin/New York, De Gruyter.
MITHUN, M. 1999. The Languages of Native North America. Cambridge, Cambridge University Press.
MONTLER, T. 1986. An Outline of the Morphology and Phonology of Saanich, North Straits Salish. Missoula, MT, University of Montana Occasional Papers in Linguistics No. 4.
MOTTAUSCH, K.-H. 2000. Die Vorgeschichte des indogermanischen Vokalsystems. Ein Versuch. IF 105, 68–100.
NASSIVERA, M. 2000. The development of the PIE words for 'sky', 'cow' and 'ship' and the relative chronology of Osthoff's law. HS 113, 57–70.
NATER, H. F. 1979. Bella Coola phonology. Lingua 49, 169–187.
NEWMAN, S. 1947. Bella Coola I: phonology. IJAL 13, 129–134.
NORMIER, R. 1977. Idg. Konsonantismus, germ. »Lautverschiebung« und Vernersches Gesetz. KZ 91, 171–218.
–, 1981. Zu Esche und Espe. Die Sprache 27, 21–29.
RASMUSSEN, J. 1990/91[1999]. Some additional examples of PIE *-$\bar{e}h_2$- and *-$h_2\bar{e}$-, in J. R., Selected Papers on Indo-European Linguistics, vol. 2, 394–412. Copenhagen, Museum Tusculanum Press (Univ. of Copenhagen).
SCHULZE, W. 1913. Gk. *σφηνόπους*. KZ 45, 190–191.
–, 1933[1966]. Kleine Schriften^2. Göttingen, Vandenhoeck & Ruprecht.
SCHWYZER *Del.*: SCHWYZER, E. 1923. Dialectorum Graecarum exempla epigraphica potiora. repr. Hildesheim, Olms.
–, 1939. Griechische Grammatik, vol. I. Munich, Beck.

SOLMSEN, F. 1909. Beiträge zur griechischen Wortforschung. Straßburg, Trübner.
WAHRMANN, P. 1914. *σφαλός, σφάνιον*. Glotta 6, 162–164.
WATKINS, C. 2000. The American Heritage Dictionary of Indo-European Roots^2. Boston/New York, Houghton Mifflin.
WD: [WACKERNAGEL, J. and] A. DEBRUNNER. 1954. Altindische Grammatik, II.2 (Die Nominalsuffixe). Göttingen, Vandenhoeck & Ruprecht.